Department of
Computer Science and Artificial Intelligence

BIG DATA MANAGEMENT

Project Report

RESORT MANAGEMENT DATA WAREHOUSE AND BI
(Data Warehouse Design and Business Intelligence Implementation)

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Abstract

This project presents the design and implementation of a Data Warehouse and Business Intelligence solution for a Resort Management System. The work was carried out using the Entity–Relationship (E/R) schema provided as the basis for the project and followed the requirements specified in the Big Data Management course.

The project began with the formulation of ten decisional queries representing the information needs of resort managers. These queries were used to identify candidate facts and guide the Data Warehouse design process. The original E/R schema was then refined through reification, resulting in a set of analytical entities suitable for dimensional modelling. From the identified facts, PAYMENT and ROOM_RENTAL were selected for implementation because they provided the broadest coverage of the decisional queries.

Conceptual modelling was performed using the Dimensional Fact Model (DFM) methodology. Attribute trees were generated and subsequently refined through pruning and grafting. Dimensions, measures, and hierarchies were identified and used to construct the final fact schemas. The logical design phase produced Snowflake Schemas for the PAYMENT and ROOM_RENTAL data marts.

The original operational database was implemented and populated using synthetic data generated with Python. The generated dataset was loaded into MySQL, and a SQL-based ETL process was developed to populate the Data Warehouse. The final warehouse was implemented in MySQL and consisted of shared dimensions and fact tables supporting the selected business processes.

To provide Business Intelligence capabilities, Metabase was connected to the Data Warehouse and used to develop dashboards and analytical reports. The resulting solution supports revenue analysis, accommodation performance analysis, customer demographic analysis, and seasonal reporting within a local offline environment.

Chapter 1: Introduction

1.1 Project Overview

Data Warehouses play an important role in supporting decision-making by integrating data from operational systems and organizing it for analytical reporting. In resort management environments, managers often require information related to revenue, customer behaviour, accommodation performance, investments, discounts, and seasonal trends.

This project focuses on the design and implementation of a Data Warehouse and Business Intelligence solution for a Resort Management System based on the E/R schema provided in the project specification. The project follows the complete Data Warehouse development process, including decisional query analysis, fact identification, E/R schema refinement, conceptual design, logical design, database implementation, Data Warehouse implementation, and Business Intelligence reporting.

Among the candidate facts identified during the design phase, PAYMENT and ROOM_RENTAL were selected for implementation. These facts were used to develop the data marts, analytical reports, and dashboards presented in this report.

1.2 Project Objectives

The objectives of the project are:

- Formulate decisional queries for the Resort Management System.
- Identify candidate facts from the decisional queries.
- Refine the original E/R schema through reification.
- Construct attribute trees for the selected facts.
- Apply pruning and grafting to obtain analytical structures.
- Identify dimensions, measures, and hierarchies.
- Design the final fact schemas.
- Develop Snowflake Schemas for the selected data marts.
- Implement and populate the Operational Database.
- Implement the Data Warehouse using SQL-based ETL processes.
- Develop dashboards and analytical reports using Metabase.

1.3 Tools and Technologies Used

Table 1.1 summarizes the tools and technologies used throughout the project.

Table 1.1: Tools and Technologies Used

Tool/Technology	Purpose
Draw.io	Creation of reified schemas, attribute trees, pruned and grafted trees, fact schemas, and snowflake schemas
Python	Synthetic dataset generation
MySQL	Operational Database and Data Warehouse implementation
SQL	Database creation, population, and ETL processing
MySQL Workbench	Database development and administration
Metabase	Dashboard development and Business Intelligence reporting

All components of the project were implemented in a local offline environment. Metabase was deployed using a self-hosted Java JAR installation and connected directly to the Data Warehouse.

1.4 Report Structure

The remainder of this report is organized as follows:

- **Chapter 2** presents the requirements analysis, original E/R schema, decisional queries, and candidate facts.
- **Chapter 3** presents the reification process and fact selection.
- **Chapter 4** presents the conceptual design, including attribute trees, pruning and grafting, dimensions, measures, and fact schemas.
- **Chapter 5** presents the logical design of the PAYMENT and ROOM_RENTAL data marts.
- **Chapter 6** presents the implementation and population of the Operational Database.
- **Chapter 7** presents the Data Warehouse implementation and ETL process.
- **Chapter 8** presents the Business Intelligence dashboards and analytical reports.
- **Chapter 9** discusses the analytical results obtained from the implemented solution.
- **Chapter 10** concludes the project and outlines possible future enhancements.

Chapter 2: Requirements Analysis

2.1 Overview

The first stage of the project focused on analysing the Resort Management System and identifying the information requirements that the Data Warehouse should support. This was achieved by examining the original E/R schema and formulating decisional queries that represent the information needs of resort managers.

The decisional queries served as the basis for identifying candidate facts and guided the subsequent conceptual and logical design activities.

2.2 Original E/R Schema

The project was based on the Resort Management System E/R schema provided in the assignment specification. The schema models the main operational activities of the resort, including customer payments, ROOM_RENTALs, services, investments, discounts, social media interactions, and stock market activities.

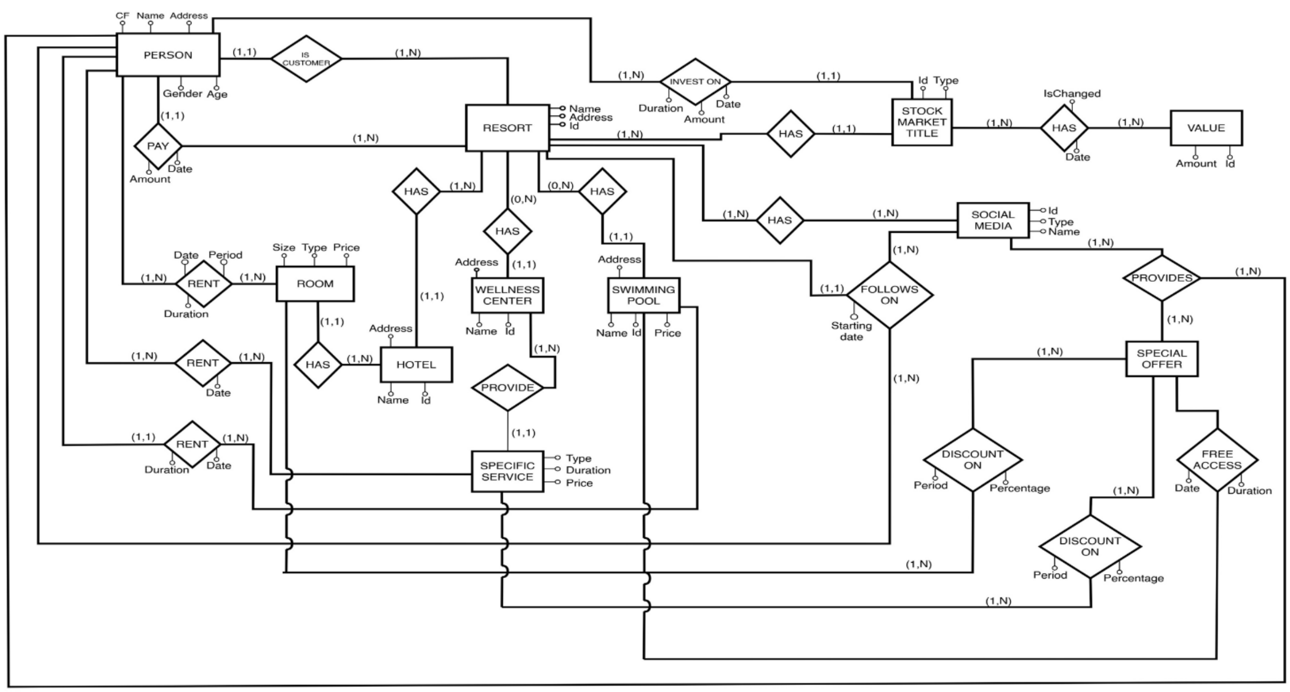


Figure 2.1: Original E/R Schema

The original schema provided the foundation for identifying business processes that could support analytical reporting. These processes were later examined through decisional queries to determine the most suitable candidate facts for the Data Warehouse.

2.3 Decisional Queries

To identify the analytical requirements of the Resort Management System, ten decisional queries were formulated. These queries represent the type of information that managers may require when analysing resort operations and business performance.

Table 2.1: Decisional Queries

No.	Query
1	What is the total and average revenue generated by customers from Italy during the month of August 2025?
2	During the Summer Business Season of 2025, which resort achieved the highest total ROOM_RENTAL revenue, and which room type contributed most to that revenue?
3	What is the total rental income generated by Suite room types within Hotel Alpha during the Summer Business Season of 2025?
4	What is the total amount invested in Technology sector stock market titles by guests from France during 2025?
5	What is the count of new followers acquired on the Instagram platform during January 2025 from Italy?
6	What is the average discount percentage applied to Family ROOM_RENTALs during the Summer Business Season of 2025 in Hotel Europa?
7	What is the usage frequency and total revenue generated by the Massage service type for each month of 2025?
8	What is the average payment amount for Male customers in the 65+ Age Group from Italy during 2025?
9	What is the occupancy rate or rental utilization level for Double room types in Hotel Beta during the Winter Business Season of 2025?
10	How many value changes did AppleStock experience, and what was the average value variation per month during 2025?

The queries cover multiple business areas.

2.4 Facts Derived from Decisional Queries

Each decisional query was analysed to determine the business event responsible for generating the required information. The resulting candidate facts are summarized in Table 2.2.

Table 2.2: Facts Derived from Decisional Queries

Query	Business Area	Main Fact
Q1	Revenue by customer geography/time	PAY
Q2	ROOM_RENTAL Performance Analysis	RENT
Q3	Suite ROOM_RENTAL income	RENT
Q4	Investment analysis	INVEST ON
Q5	Social media followers	FOLLOWS ON
Q6	Discount analysis	DISCOUNT ON
Q7	Service revenue	RENT
Q8	Demographic payment analysis	PAY
Q9	Room utilization	RENT
Q10	Stock value changes	HAS

Table 2.2 shows the relationships support multiple decisional queries and represent the most significant business processes within the Resort Management System.

2.5 Facts Selection

The facts identified from the decisional queries were evaluated based on their analytical relevance and coverage of the business requirements.
The PAY relationship supports revenue analysis, geographical analysis, and demographic analysis through Queries Q1 and Q8.
The RENT relationship supports accommodation performance analysis, service revenue analysis, room utilization analysis, and rental income analysis through Queries Q2, Q3, Q7, and Q9.
Although other candidate facts were identified, PAY and RENT provided the broadest coverage of the decisional queries and were therefore selected for implementation in the Data Warehouse.
The remaining facts were retained during the conceptual design phase but were not implemented in the final Data Warehouse.

2.6 Summary

This chapter analysed the requirements of the Resort Management System using the original E/R schema and a set of decisional queries.
The next chapter refines the original E/R schema through reification and transforms the selected business processes into analytical entities suitable for dimensional modelling.

Chapter 3: Fact Identification and E/R Refinement

3.1 Overview

Following the requirements analysis presented in Chapter 2, the original E/R schema was refined to prepare it for dimensional modelling. This refinement focused on relationships representing business events that could support analytical reporting.
The refinement was performed through reification, resulting in a set of analytical entities that could be used during the conceptual design phase.

3.2 Reification Process

The candidate facts identified in Chapter 2 were originally represented as relationships in the operational E/R schema. To support dimensional modelling, these relationships were reified into analytical entities.
During reification, relationship attributes were transferred to the corresponding reified entity. New relationships were then introduced between the reified entity and the participating entities.

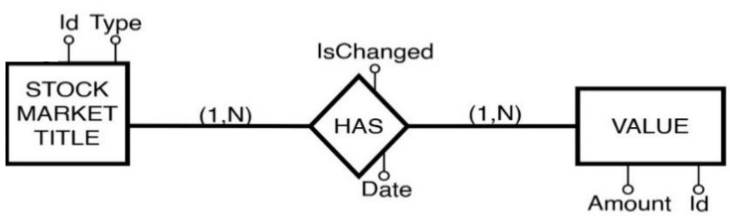


Figure 3.1: Original HAS Relationship

Figure 3.1 shows the HAS relationship before reification. The relationship contains the attributes Date and IsChanged and represents an association between STOCK MARKET TITLE and VALUE.

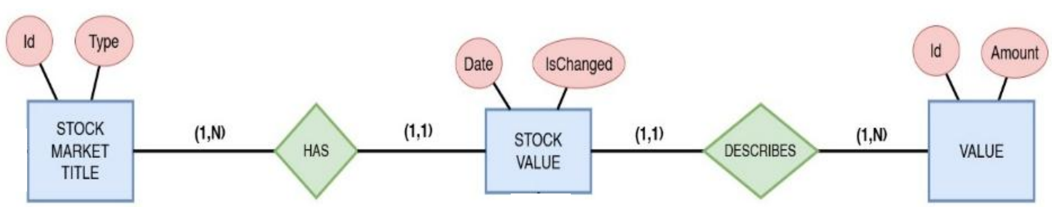


Figure 3.2: Reified HAS Relationship

After reification, the HAS relationship was transformed into the STOCK VALUE entity. The attributes Date and IsChanged became attributes of STOCK VALUE.
The original HAS relationship with cardinalities (1,N) and (1,N) was replaced by the STOCK VALUE entity. The new relationships, HAS and DESCRIBES, introduce a cardinality of (1,1) on the STOCK VALUE side, preserving the original business constraints represented in the E/R schema.
The same procedure was applied to all selected relationships in the schema.

3.3 Reified Entities

Table 3.1 summarizes the relationships selected for reification and their corresponding analytical entities.

Table 3.1: Reification of Relationships into Analytical Entities

Relationship	Reified Entity
PAY	PAYMENT
RENT	ROOM_RENTAL
RENT	SERVICE RENTAL
RENT	POOL RENTAL
INVEST ON	INVESTMENT
FOLLOWS ON	FOLLOW
DISCOUNT ON	DISCOUNT ROOM
DISCOUNT ON	DISCOUNT SERVICE
HAS	STOCK VALUE

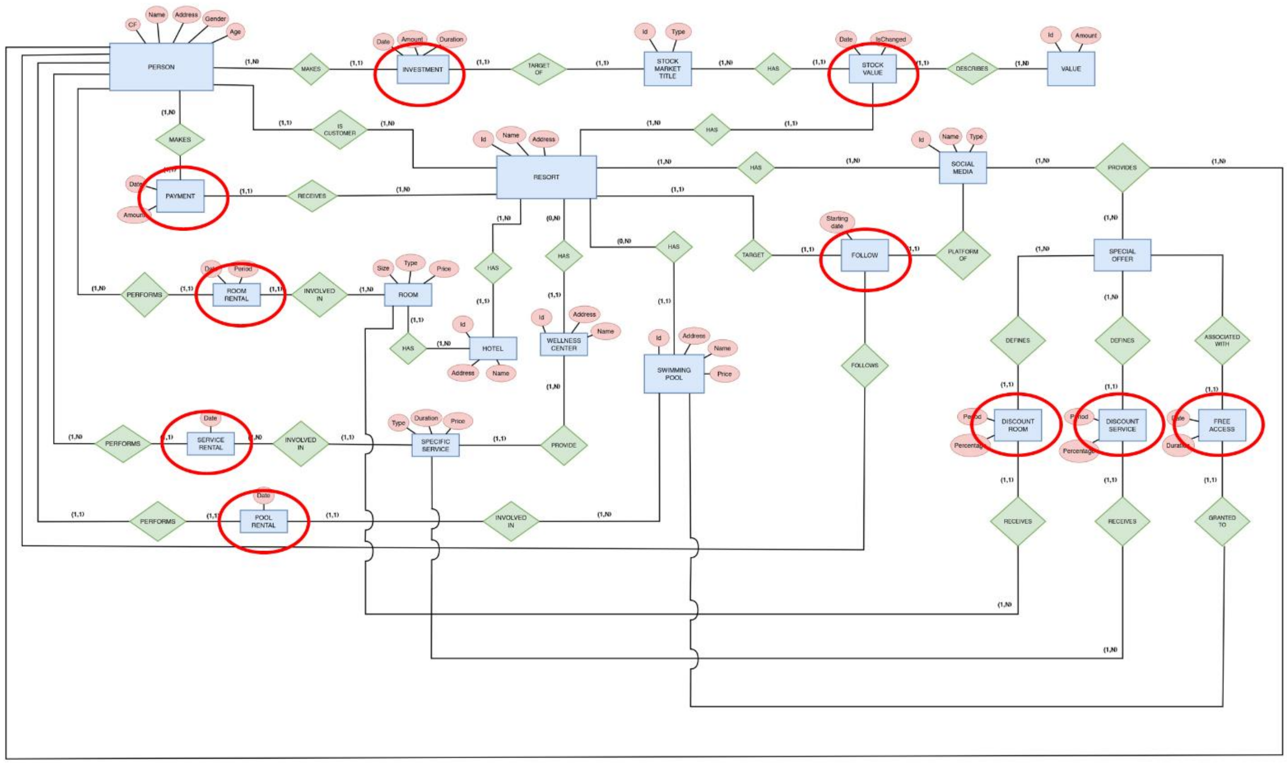


Figure 3.3: Reified E/R Schema

The reified schema provides a clearer representation of the business events contained within the Resort Management System and forms the basis for conceptual modelling.

3.4 Fact Selection

The reified entities were evaluated against the decisional queries identified in Chapter 2.

PAYMENT supports:

- Q1 Revenue by customer geography and time
- Q8 Demographic payment analysis

ROOM_RENTAL supports:

- Q2 ROOM_RENTAL performance analysis
- Q3 Suite ROOM_RENTAL income
- Q7 Service revenue analysis
- Q9 Room utilization analysis

Since PAYMENT and ROOM_RENTAL support the largest number of decisional queries, they were selected for implementation in the Data Warehouse.

3.5 Summary

This chapter refined the original E/R schema through reification and produced the analytical entities required for dimensional modelling. Based on the decisional queries, PAYMENT and ROOM_RENTAL were selected for implementation. The next chapter develops the conceptual design through attribute trees, pruning and grafting, dimensions, measures, and fact schemas.

Chapter 4: Conceptual Design

4.1 Overview

Conceptual modelling was performed for all reified entities identified in Chapter 3. Attribute trees were generated automatically and subsequently refined through pruning and grafting to identify dimensions, measures, and hierarchies suitable for dimensional modelling.

For reporting purposes, this chapter presents only the conceptual design of the PAYMENT and ROOM_RENTAL facts, which were selected for implementation in the Data Warehouse. The same modelling procedure was applied to the remaining candidate facts during the design process.

4.2 Pruning and Grafting

The attribute trees were refined through pruning and grafting. Pruning removed attributes that do not contribute to analytical reporting, while grafting reorganized and derived attributes required by the decisional queries. Both operations were performed together to ensure that the resulting structures were suitable for dimensional modelling and aligned with the analytical requirements.

4.3 PAYMENT Fact

4.3.1 PAYMENT Attribute Tree

Figure 4.1 shows the automatically generated attribute tree for the PAYMENT fact.

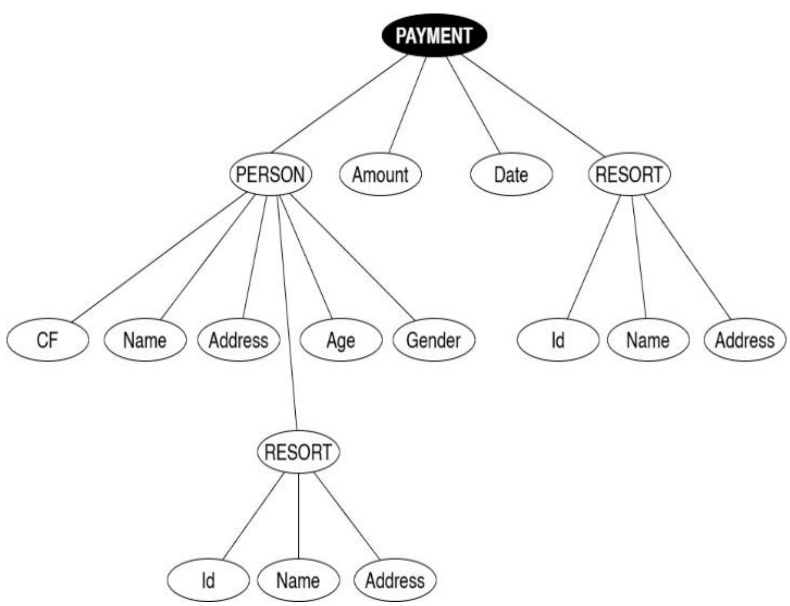


Figure 4.1: PAYMENT Attribute Tree

4.3.2 PAYMENT Pruned and Grafted Tree

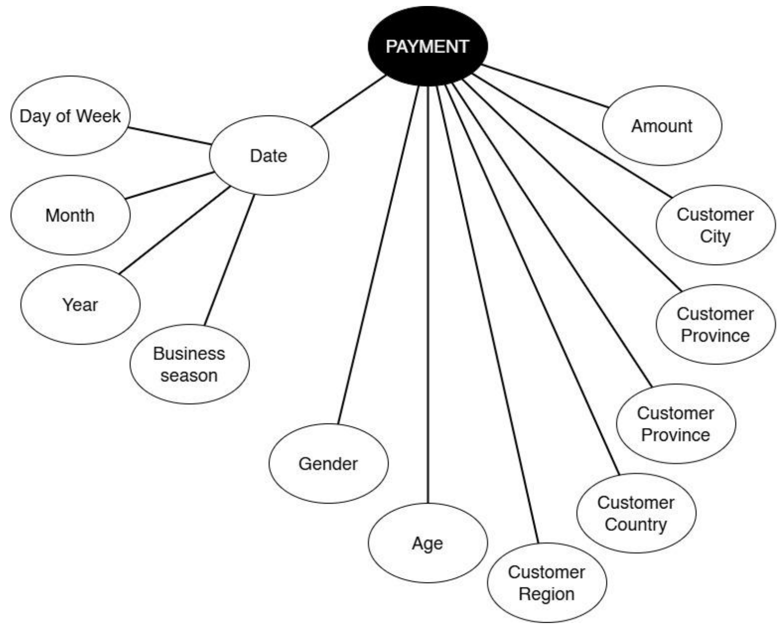


Figure 4.2: PAYMENT Pruned and Grafted Tree

The PAYMENT attribute tree was refined to support revenue analysis across time, geography, and customer demographics.

Table 4.1: PAYMENT Prunned and Grafted Attributes

Prunned Atrributes	**Grafted Attributes**
CF	Day of Week
Customer Name	Month
Customer Address	Year
Resort ID	Business Season
Resort Name	Customer City
Resort Address	Customer Province
	Customer Region
	Customer Country

4.3.3 PAYMENT Dimensions and Measures

Dimensions and measures were selected from the pruned and grafted tree based on the requirements of Q1 and Q8 as shown in Table 4.2.

Table 4.2: PAYMENT Dimensions and Measures

Measures	**Dimensions**
SUM(Amount) AVG(Amount) COUNT(PAY)	**Time** • Date • Day of Week • Month • Year • Business Season **Customer Geography** • Customer City • Customer Province • Customer Country **Customer Demographics** • Gender • Age Group **Resort** • Region

4.3.4 PAYMENT Fact Schema

Figure 4.3 presents the final fact schema for PAYMENT.

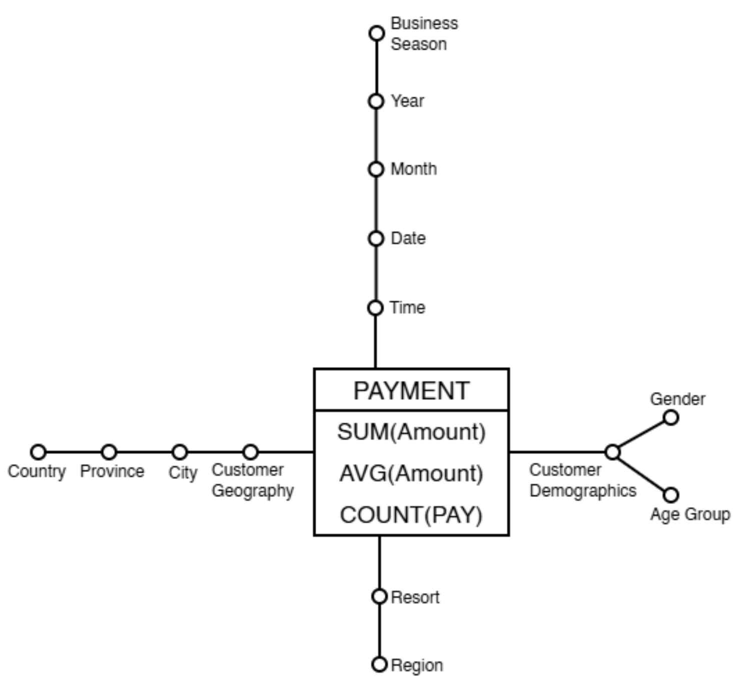


Figure 4.3: PAYMENT Fact Schema

4.4 ROOM_RENTAL Fact

4.4.1 ROOM_RENTAL Attribute Tree

Figure 4.4 shows the automatically generated attribute tree for the ROOM_RENTAL fact.

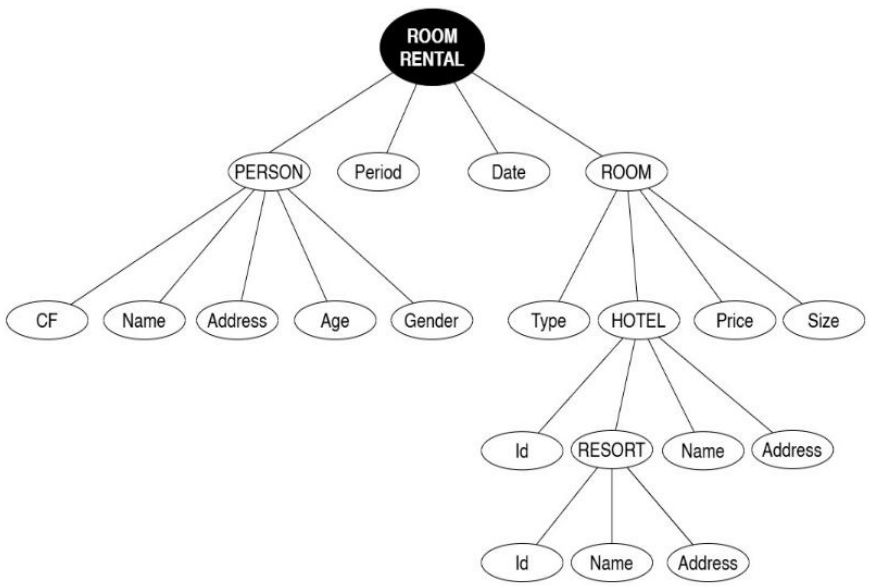


Figure 4.4: ROOM_RENTAL Attribute Tree

4.4.2 ROOM_RENTAL Pruned and Grafted Tree

The ROOM_RENTAL attribute tree was refined to support accommodation performance and room utilization analysis.

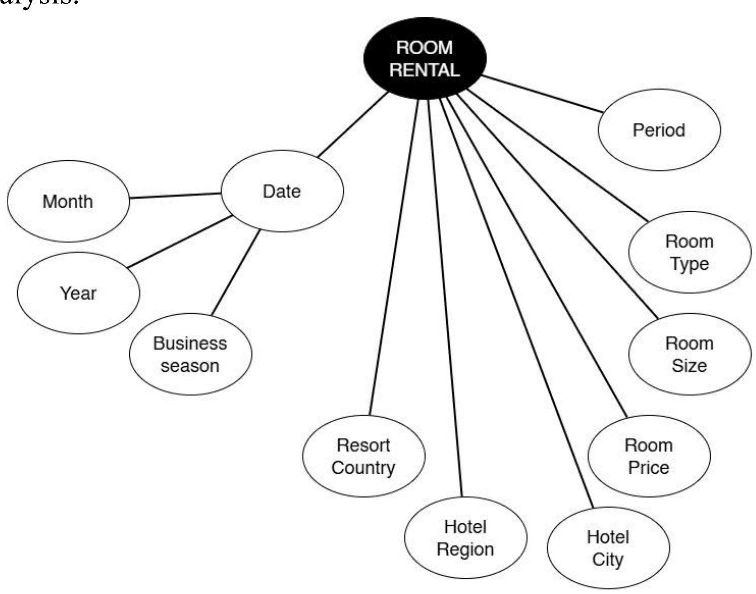


Figure 4.5: ROOM_RENTAL Pruned and Grafted Tree

4.4.3 ROOM_RENTAL Dimensions and Measures

Dimensions and measures were selected from the pruned and grafted tree based on the requirements of Q2, Q3, Q7, and Q9.

Table 4.2: ROOM_RENTAL Dimensions and Measures

Measures	**Dimensions**
SUM(Room Price) AVG(Room Price) COUNT(RENT) AVG(Period)	**Time** • Date • Month • Year • Business Season **Room** • Room Type • Room Size **Hotel/Resort** • City • Region • Country

4.4.4 ROOM_RENTAL Fact Schema

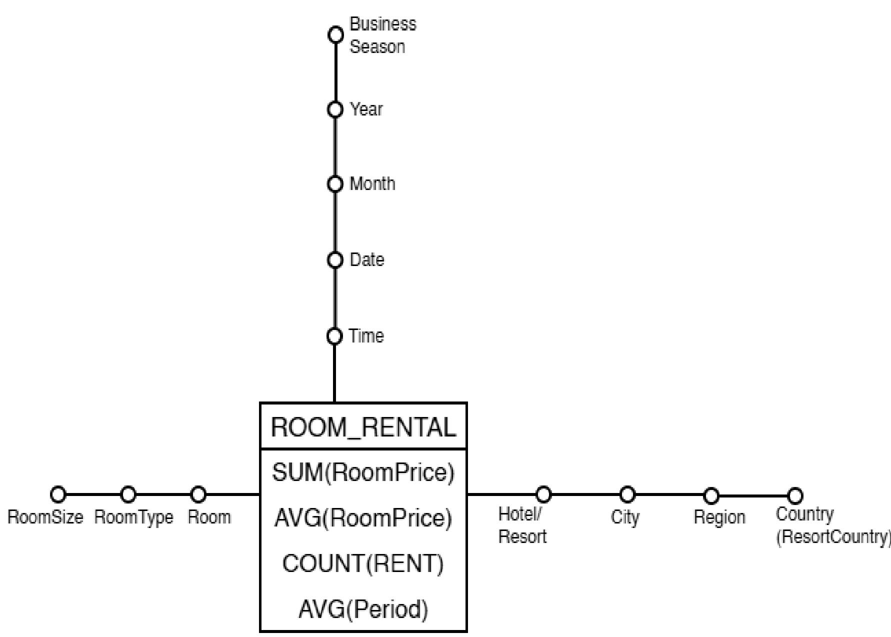


Figure 4.6: ROOM_RENTAL Fact Schema

4.5 Summary

This chapter presented the conceptual design of the PAYMENT and ROOM_RENTAL facts. Attribute trees were generated and refined through pruning and grafting, resulting in analytical structures suitable for dimensional modelling. The resulting dimensions, measures, and hierarchies were then used to construct the final fact schemas. The next chapter presents the logical design of the corresponding data marts.

Chapter 5: Logical Design

5.1 Overview

The logical design phase transformed the conceptual models developed in Chapter 4 into data marts suitable for Data Warehouse implementation. The identified dimensions, measures, and hierarchies were used to develop snowflake schemas for the PAYMENT and ROOM_RENTAL facts.

5.2 Hierarchy Definition

The hierarchies were derived from the dimensions identified during conceptual design. These hierarchies support drill-down and roll-up operations during multidimensional analysis.

Table 5.1: Selected Hierarchies

Hierarchy	Levels
Time	Date → Month → Year
Customer Geography	City → Province → Country
Resort Geography	Resort → Region
Room	Room Type → Room Size
Hotel Geography	Hotel → City → Region → Country

5.2.1 Key Definition and Relationship Structure

Primary keys and foreign keys were identified during the logical design phase to support the implementation of the snowflake schemas.

- **Primary keys** were assigned to uniquely identify records within each dimension and fact table.

- **Foreign keys** were introduced to represent the relationships between hierarchy levels and to maintain referential integrity across the data marts.

- The hierarchy relationships identified during conceptual design were translated into linked dimension tables. For example, the Customer Geography hierarchy was implemented through the relationships: City → Province → Country

- Similarly, the Resort Geography, Hotel Geography, and Room hierarchies were implemented using foreign key relationships between successive hierarchy levels.

5.3 PAYMENT Data Mart

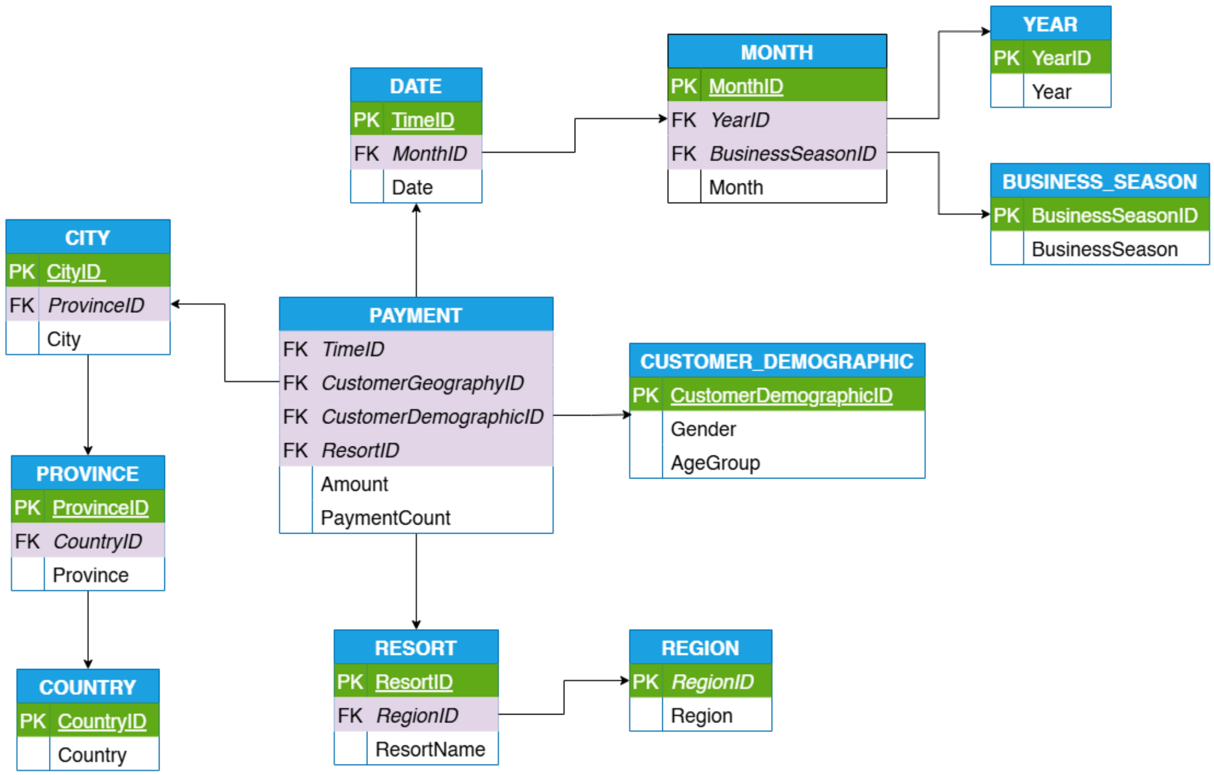


Figure 5.1: PAYMENT Snowflake Schema

The schema consists of the PAYMENT fact table and the associated dimensions for Time, Customer Geography, Customer Demographic, and Resort. Hierarchical relationships were normalized into separate dimension tables to support geographical and temporal analysis at multiple levels of aggregation.

5.4 ROOM_RENTAL Data Mart

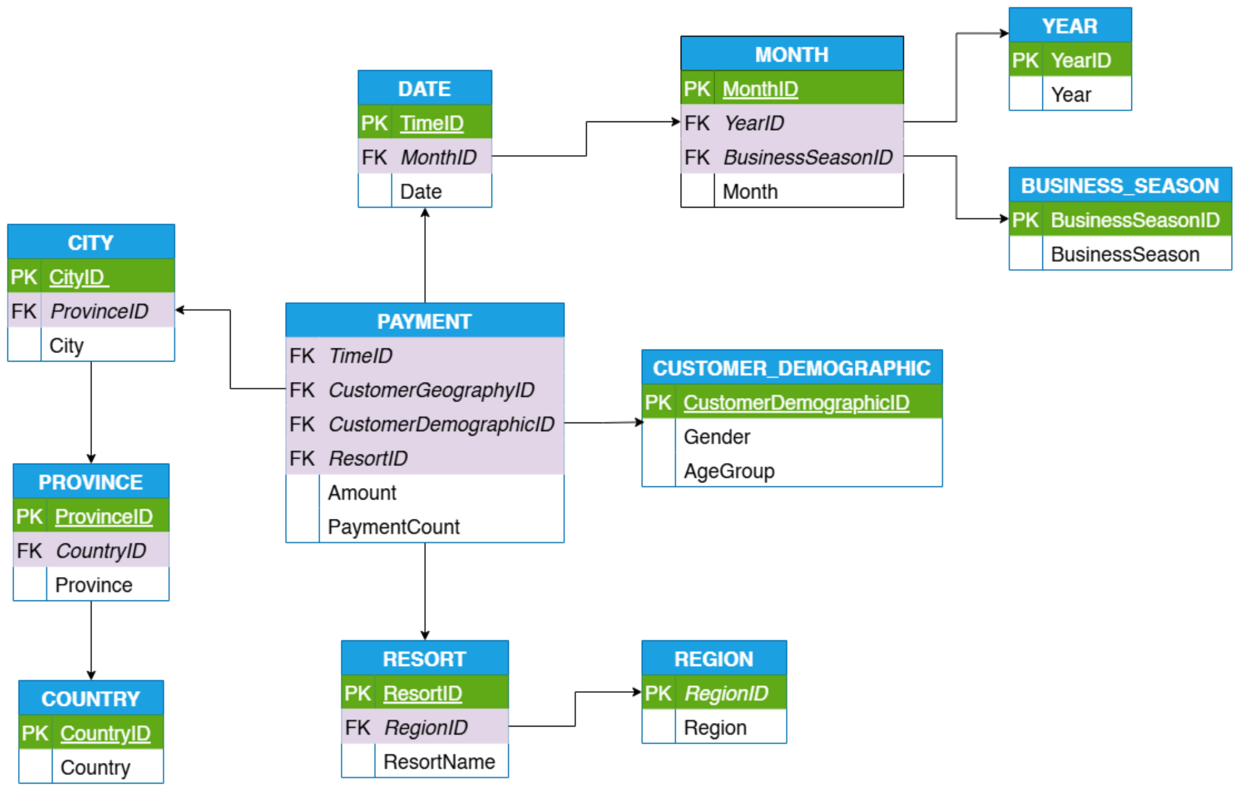


Figure 5.2: ROOM_RENTAL Snowflake Schema

The ROOM_RENTAL data mart was designed using a snowflake schema. The schema implements the dimensions, measures, and hierarchies identified during conceptual design and supports accommodation performance, rental revenue, and room utilization analysis.

5.5 Summary

This chapter presented the logical design of the PAYMENT and ROOM_RENTAL data marts. The selected hierarchies were used to construct snowflake schemas that form the basis for the Data Warehouse implementation presented in the next chapter.

Chapter 6: Operational Database Implementation

6.1 Overview

This chapter presents the implementation and population of the operational database based on the reified E/R schema developed in Chapter 3. The operational database was implemented in MySQL and populated using synthetic data generated with Python.

The implementation process consisted of three main stages:

- Dataset generation
- Database creation
- Database population

The resulting operational database served as the source system for the Data Warehouse implementation presented in Chapter 7.

6.2 Operational Database Design

The operational database was implemented from the reified E/R schema presented in Chapter 3. Figure 6.1 presents the implemented operational database schema which contains the primary entities; PERSON, RESORT, HOTEL, ROOM, PAYMENT, ROOM_RENTAL, SERVICE_RENTAL, POOL_RENTAL, INVESTMENT, FOLLOW, DISCOUNT_ROOM, DISCOUNT_SERVICE, STOCK_VALUE

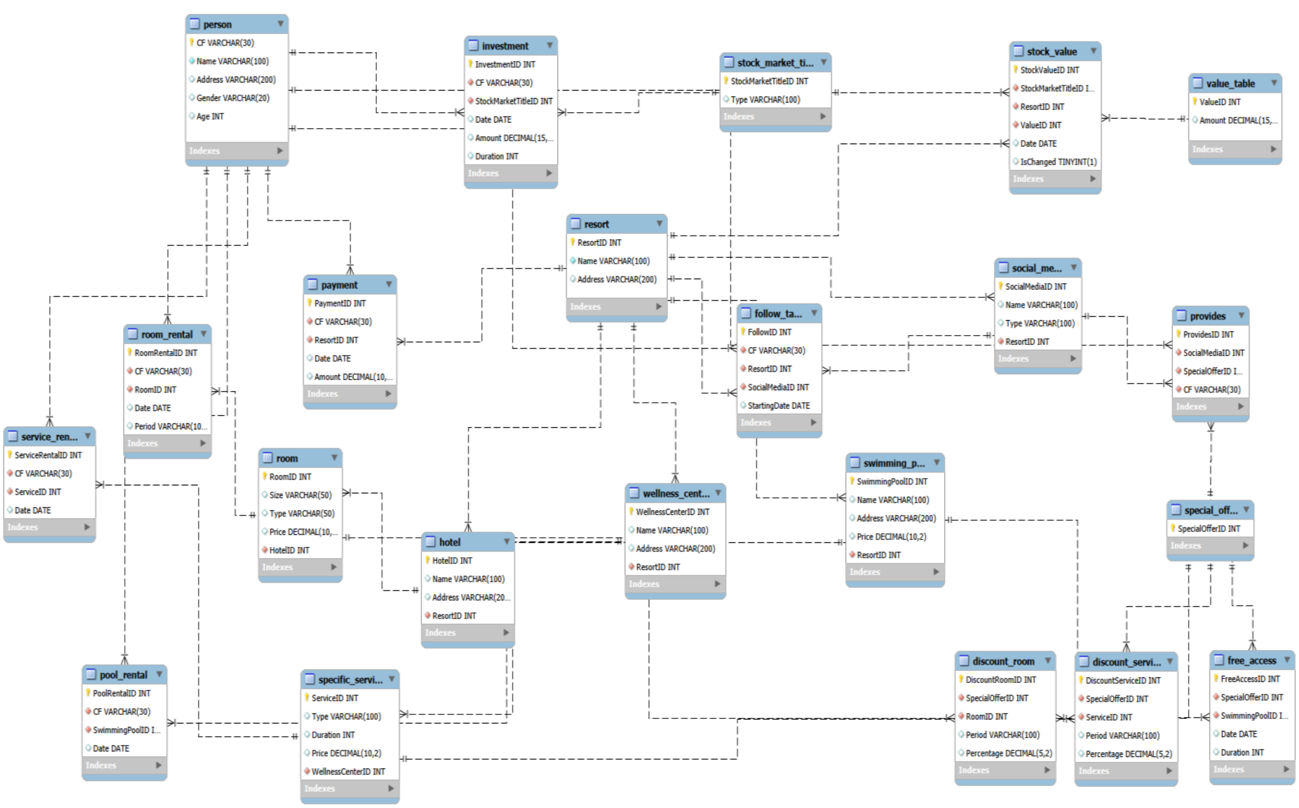


Figure 6.1: Operational Database Schema

6.3 Dataset Generation

Synthetic data was generated using Python to populate the operational database. The generated data was designed to simulate realistic resort operations while maintaining consistency with the implemented schema.

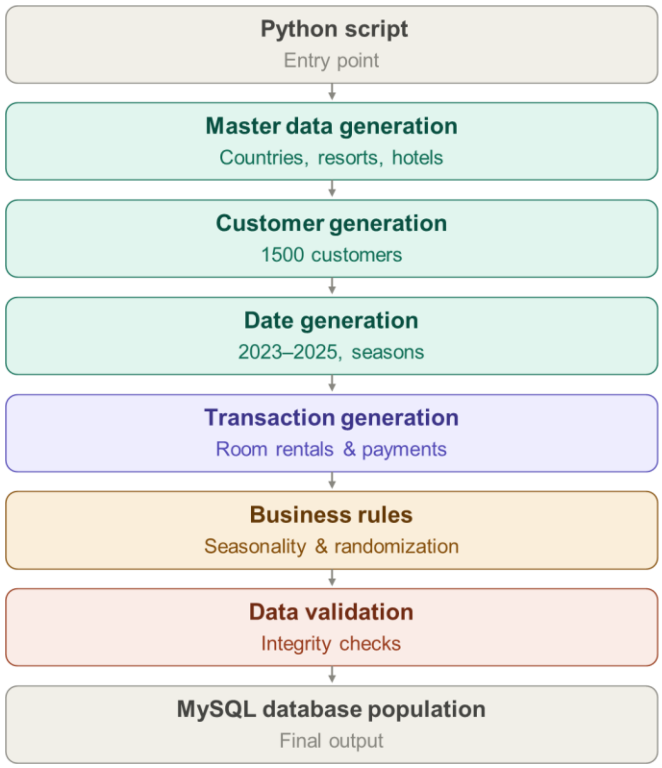


Figure 6.2: Python Data Generation Flowchart

6.3.1 Python Libraries Used

Table 6.1: Python Libraries Used

Library	Usage
Random (python library)	Generation of randomized business data
Datetime (python library)	Creation and management of transaction dates
collections.Counter (python library)	Validation and analysis of generated data distributions

The Python libraries listed in Table 6.1 were used to generate random records, create dates, and validate generated data distributions.

```python
import random
from datetime import date, timedelta
from collections import Counter

```

Code Listing 6.3: Python Libraries Used

6.3.2 Dataset Generation Parameters

The dataset generation process defined the number of records required for each entity in the operational database.

Table 6.2: Generated Dataset Volumes

Entity	**Records Generated**
PERSON	1,500
RESORT	5
HOTEL	10
ROOM	100
WELLNESS_CENTER	8
SPECIFIC_SERVICE	35
SWIMMING_POOL	15
SOCIAL_MEDIA	15
STOCK_MARKET_TITLE	10
VALUE_TABLE	150
STOCK_VALUE	250
SPECIAL_OFFER	60
ROOM_RENTAL	900
SERVICE_RENTAL	500
POOL_RENTAL	300
INVESTMENT	250
FOLLOW	600
DISCOUNT_ROOM	80
DISCOUNT_SERVICE	70
FREE_ACCESS	50
PROVIDES	300
PAYMENT	Generated separately (1–10 payments per customer)

```
import random
from datetime import date, timedelta
from collections import Counter

random.seed(42)

OUTPUT_FILE = "populate_resort_operational_db.sql"

# =====================================================
# MAIN SETTINGS
# =====================================================

PERSON_COUNT = 1500
RESORT_COUNT = 5
HOTEL_COUNT = 10
ROOM_COUNT = 100
WELLNESS_CENTER_COUNT = 8
SPECIFIC_SERVICE_COUNT = 35
SWIMMING_POOL_COUNT = 15
SOCIAL_MEDIA_COUNT = 15
STOCK_MARKET_TITLE_COUNT = 10
VALUE_TABLE_COUNT = 150
STOCK_VALUE_COUNT = 250
SPECIAL_OFFER_COUNT = 60

ROOM_RENTAL_COUNT = 900
SERVICE_RENTAL_COUNT = 500
POOL_RENTAL_COUNT = 300
INVESTMENT_COUNT = 250
FOLLOW_COUNT = 600
DISCOUNT_ROOM_COUNT = 80
DISCOUNT_SERVICE_COUNT = 70
FREE_ACCESS_COUNT = 50
PROVIDES_COUNT = 300

# PAYMENT is generated separately:
# each person makes 1 to 10 payments.
```

Code Listing 6.2: Dataset Generation Parameters

6.3.3 Seasonal Data Distribution

To simulate realistic resort activity, the generated dataset incorporated seasonal behaviour. Records were generated between 2023 and 2025, with a higher probability assigned to the summer months.

```
def weighted_tourism_date():
    """
    Generates dates between 2023 and 2025.
    Summer months have the highest probability.
    """
    year = random.choice([2023, 2024, 2025])

    month = random.choices(
        population=[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12],
        weights=[3, 3, 4, 6, 8, 14, 18, 18, 10, 7, 5, 4],
        k=1
    )[0]
```

Code Listing 6.3: Summer Season Bias Logic

The seasonal weighting increased the frequency of transactions occurring during June, July, and August.

6.3.4 Date Generation Logic

Date values used throughout the generated dataset were randomly selected within the defined operational period.

```
    year = random.choice([2023, 2024, 2025])
```

Code Listing 6.4: Date Generation Logic

The generated dates support temporal analysis across multiple years and seasonal periods.

6.4 Database Creation

The operational database was created in MySQL using SQL Data Definition Language (DDL) statements. Tables, primary keys, foreign keys, and integrity constraints were defined according to the implemented schema.

```
CREATE DATABASE IF NOT EXISTS resort_operational_db;

USE resort_operational_db;

-- ========================================================
-- 1. PERSON
-- ========================================================

CREATE TABLE PERSON (
    CF VARCHAR(30) PRIMARY KEY,
```

Code Listing 6.5: Database Creation Script Excerpt

The complete database creation script is available in the project GitHub repository (Appendix A).

6.5 Database Population

After creating the database structure, the generated dataset was loaded into the corresponding tables. The population process maintained referential integrity between related entities and relationships.

```
INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (1, 'CF00001', 1, '2024-08-31', 1430.11);
INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (2, 'CF00001', 4, '2024-09-20', 497.92);
INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (3, 'CF00001', 3, '2025-10-11', 838.83);
INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (4, 'CF00001', 3, '2025-07-20', 338.72);
INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (5, 'CF00001', 3, '2024-09-01', 522.09);
INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (6, 'CF00001', 4, '2025-09-30', 242.20);
INSERT INTO PAYMENT (PaymentID, CF, ResortID, `Date`, Amount) VALUES (7, 'CF00001', 5, '2023-10-09', 956.93);
```

Code Listing 6.6: Database Population Script Excerpt

The complete database population script is available in the project GitHub repository (Appendix A).

Successful population of the operational database provided the source data required for the extraction, transformation, and loading processes described in Chapter 7.

The workflow adopted in this project is shown below:

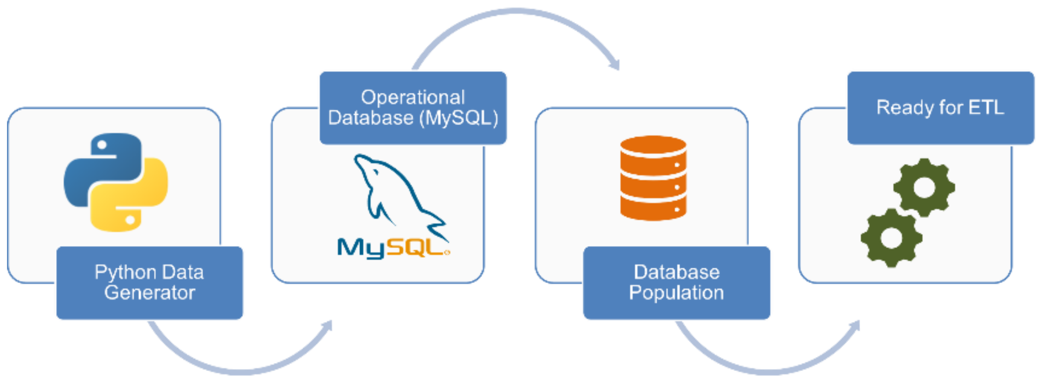


Figure 6.3: SQL Database Population Screenshot

After execution of the population script, the Operational Database contained the complete dataset required for the ETL process and subsequent Data Warehouse implementation.

6.6 Summary

This chapter presented the implementation and population of the operational database. The operational schema was implemented in MySQL and populated using synthetic data generated with Python. The resulting database provides the operational data source used for the Data Warehouse implementation presented in the next chapter.

Chapter 7: Data Warehouse Implementation

7.1 Overview

This chapter presents the implementation of the Data Warehouse and the ETL process used to transform operational data into analytical structures. The warehouse was implemented in MySQL within a dedicated database named resort_dw and supports the PAYMENT and ROOM_RENTAL business processes selected during the design phase.

7.2 Data Warehouse Workflow

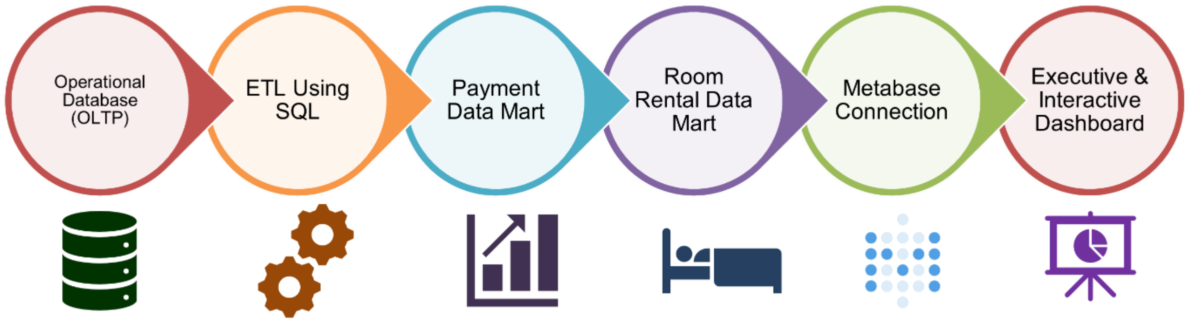


Figure 7.1: Data Warehouse Implementation Workflow

Figure 7.1 illustrates the workflow adopted in the project. Data generated and stored in the Operational Database was extracted, transformed, and loaded into the Data Warehouse using SQL scripts. The resulting warehouse served as the source for the Payment and ROOM_RENTAL Data Marts and the Business Intelligence environment implemented in Metabase.

The entire solution was deployed in a local offline environment, including the Operational Database, Data Warehouse, ETL scripts, and self-hosted Metabase installation.

7.3 Data Warehouse Architecture

The implemented architecture consists of the Operational Database, SQL-based ETL layer, Data Warehouse (resort_dw), analytical Data Marts, and the Metabase reporting layer.

The architecture follows established Data Warehouse principles separating operational and analytical processing (Golfarelli & Rizzi, 2009).

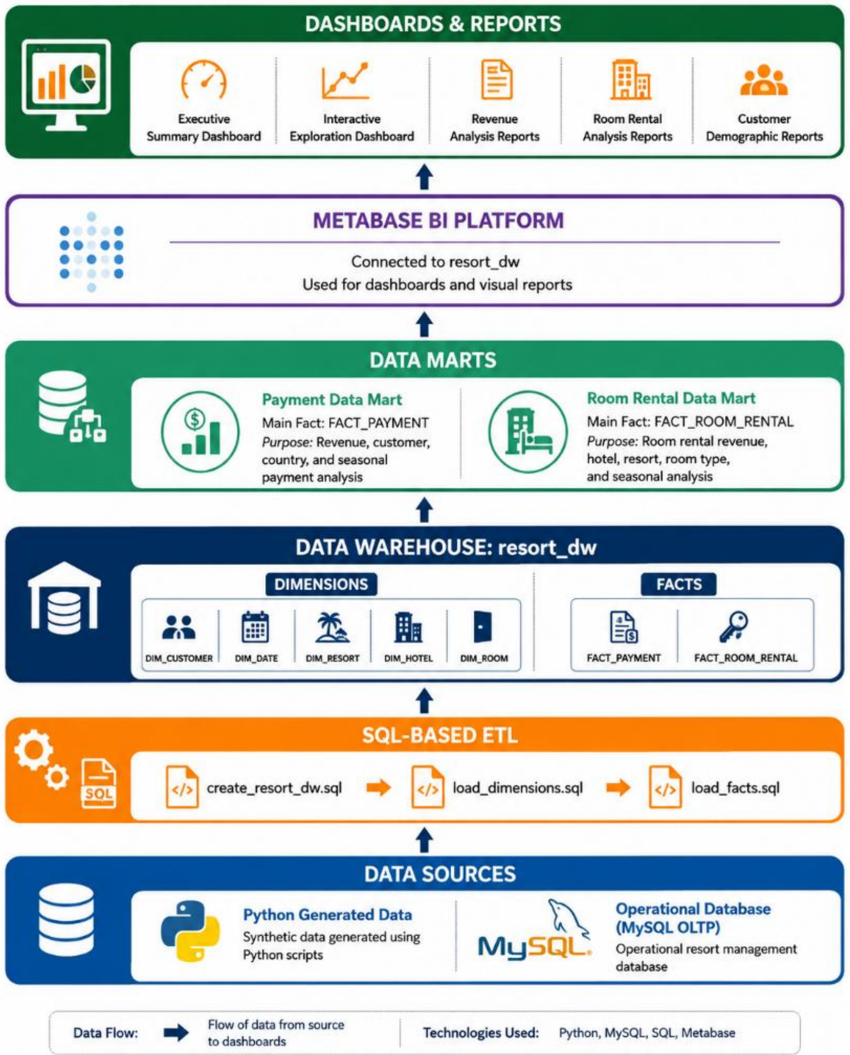


Figure 7.2: Data Warehouse Architecture

The warehouse contains five dimension tables:

- DIM_DATE
- DIM_CUSTOMER
- DIM_RESORT
- DIM_HOTEL
- DIM_ROOM

and two fact tables:

- FACT_PAYMENT
- FACT_ROOM_RENTAL

7.4 Data Warehouse Creation

The Data Warehouse was created using the create_resort_dw.sql script.

```
DROP DATABASE IF EXISTS resort_dw;
CREATE DATABASE resort_dw;
USE resort_dw;

CREATE TABLE DIM_DATE (
    DateID INT AUTO_INCREMENT PRIMARY KEY,
    FullDate DATE NOT NULL UNIQUE,
```

Code Listing 7.1: Data Warehouse Creation Script Excerpt

The script created the resort_dw database together with the required dimension and fact tables defined during the logical design phase.

7.5 ETL Implementation

ETL processes are responsible for extracting data from operational systems, transforming it into analytical structures, and loading it into a Data Warehouse (Inmon, 2005).

The ETL process was implemented using SQL scripts executed in the following sequence:

1. create_resort_dw.sql
2. load_dimensions.sql
3. load_facts.sql

This sequence ensured that all dimension records were available before loading fact records.

7.5.1 Dimension Loading

The load_dimensions.sql script populated the dimension tables from the Operational Database.

```
USE resort_dw;

INSERT INTO DIM_DATE
(FullDate, DayOfWeek, MonthNumber, MonthName, QuarterNumber, YearNumber, BusinessSeason)
SELECT DISTINCT
    d.FullDate,
    DAYNAME(d.FullDate),
    MONTH(d.FullDate),
```

Code Listing 7.2: Dimension Loading Script Excerpt

Table 7.1: Key transformations performed during dimension loading

DIM_DATE	DIM_CUSTOMER	DIM_ROOM
Month Name	Age Group	Room Type
Business Season	City	Room Size
Day of Week	Country	Room Price
Quarter Number		
Month Number		
Year Number		

7.5.2 Fact Loading

The load_facts.sql script populated FACT_PAYMENT and FACT_ROOM_RENTAL.

```sql
USE resort_dw;

INSERT INTO FACT_PAYMENT
(SourcePaymentID, DateID, CustomerDimID, ResortDimID, Amount, PaymentCount)
SELECT
    p.PaymentID,
    dd.DateID,
    dc.CustomerDimID,
    dr.ResortDimID,
    p.Amount,
```

Code Listing 7.3: Fact Loading Script Excerpt

- FACT_PAYMENT stores payment transactions together with the Amount measure and PaymentCount indicator.

- FACT_ROOM_RENTAL stores rental transactions together with RentalDays and RentalRevenue measures.

RentalRevenue was calculated as:
Rental Revenue = Room Price × Rental Days
where RentalDays was derived from the rental period stored in the Operational Database.

7.6 Implemented Data Marts

Payment Data Mart

The Payment Data Mart is centred on FACT_PAYMENT and supports:
- Revenue Analysis
- Customer Demographic Analysis
- Geographic Analysis
- Seasonal Payment Analysis

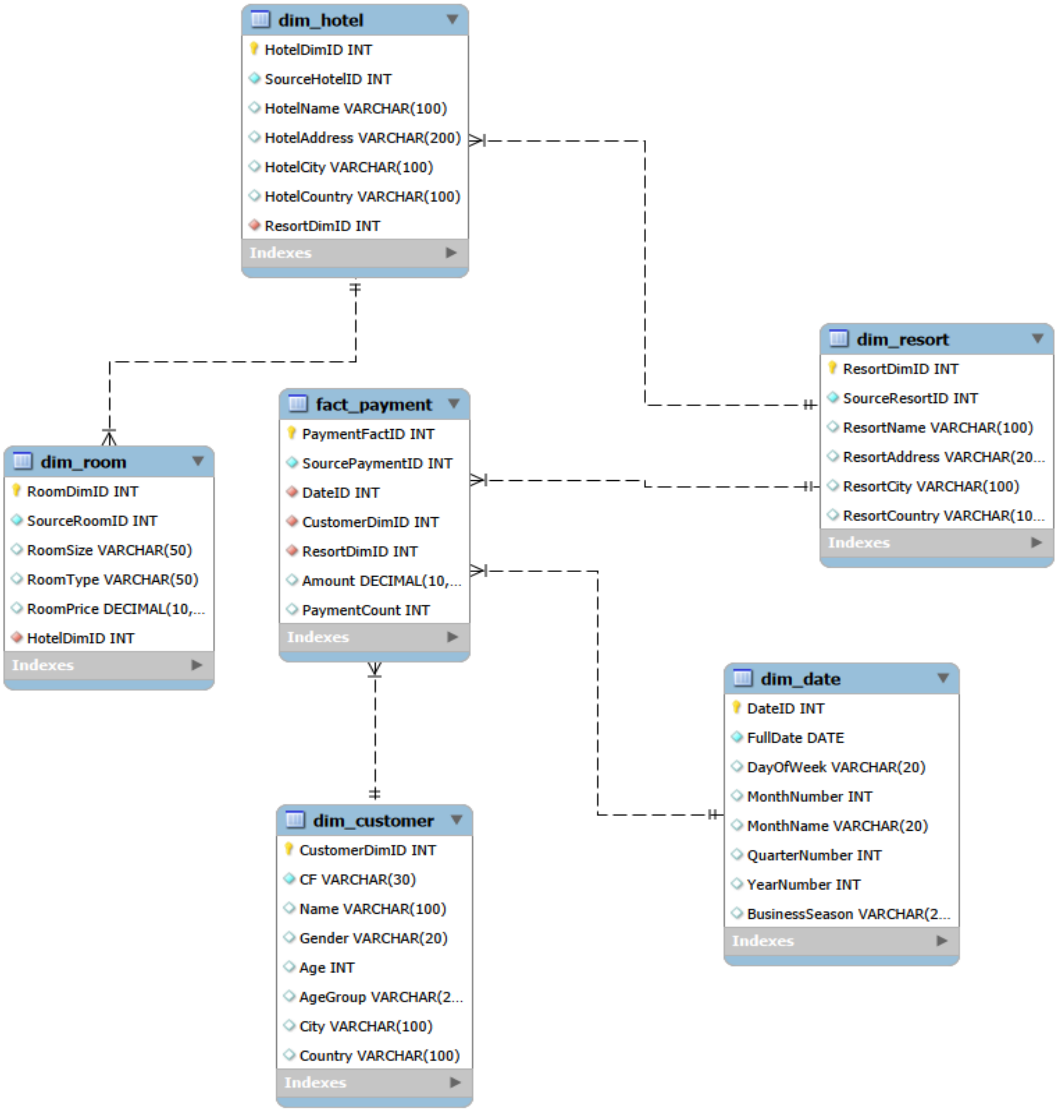


Figure 7.3: Payment Data Mart

ROOM_RENTAL Data Mart

The ROOM_RENTAL Data Mart is centred on FACT_ROOM_RENTAL and supports:

- Accommodation Performance Analysis
- Occupancy Analysis
- Hotel Performance Analysis
- Resort Performance Analysis
- Seasonal Rental Analysis

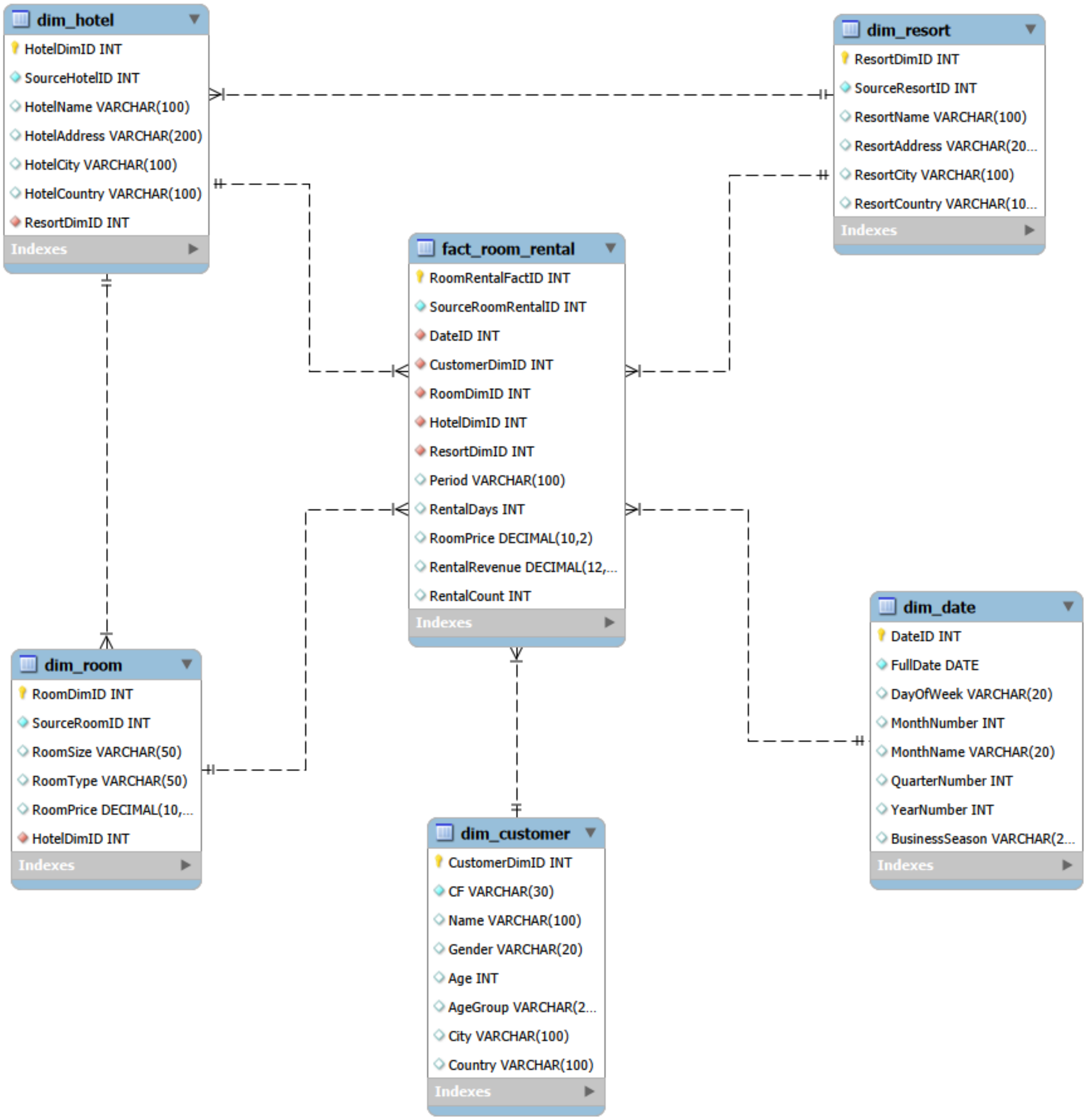


Figure 7.4: ROOM_RENTAL Data Mart

7.7 Implemented Data Warehouse Schema

Figure 7.5 presents the implemented Data Warehouse schema after completion of the ETL process. The schema integrates the dimension and fact tables required for multidimensional analysis and Business Intelligence reporting.

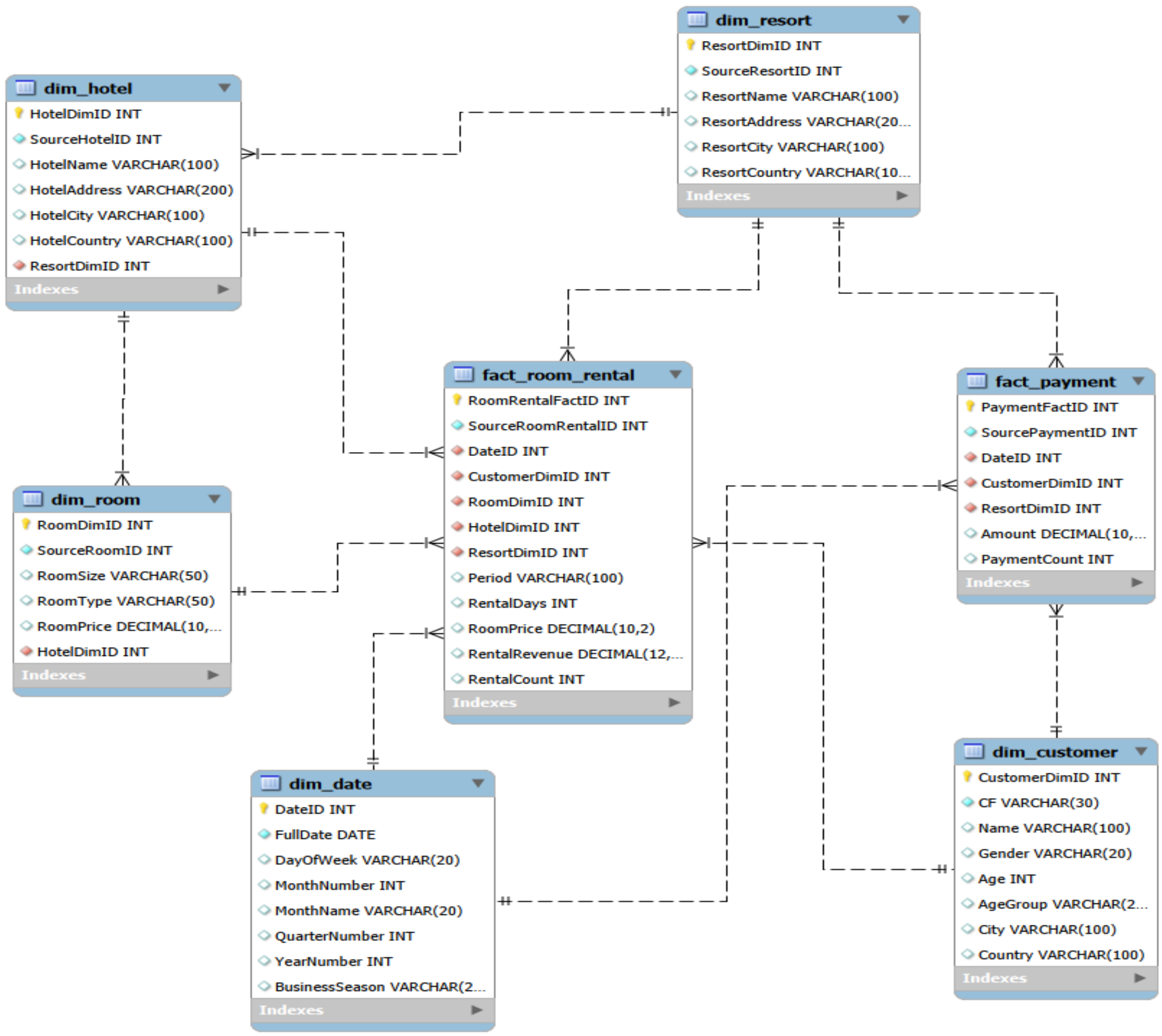


Figure 7.5: Data Warehouse Schema

7.8 Summary

This chapter presented the implementation of the Data Warehouse and the SQL-based ETL process used to populate it. The resulting warehouse consists of five dimension tables and two fact tables supporting the Payment and ROOM_RENTAL Data Marts. The implemented warehouse serves as the analytical foundation for the Business Intelligence environment presented in the next chapter.

Chapter 8: Business Intelligence and Reporting

8.1 Overview

This chapter presents the Business Intelligence environment developed using Metabase. The platform was connected to the implemented Data Warehouse (resort_dw) and used to create dashboards and analytical reports that support the decisional queries identified during the requirements analysis phase.
The Business Intelligence layer provides KPI monitoring, geographical analysis, demographic analysis, seasonal analysis, and ROOM_RENTAL performance analysis through interactive visualizations and reports (Metabase, n.d.).

8.2 Metabase Configuration

Metabase was deployed locally using the Java JAR installation and connected directly to the resort_dw Data Warehouse.

Figure 8.1 shows the configuration used to connect Metabase to the Data Warehouse.

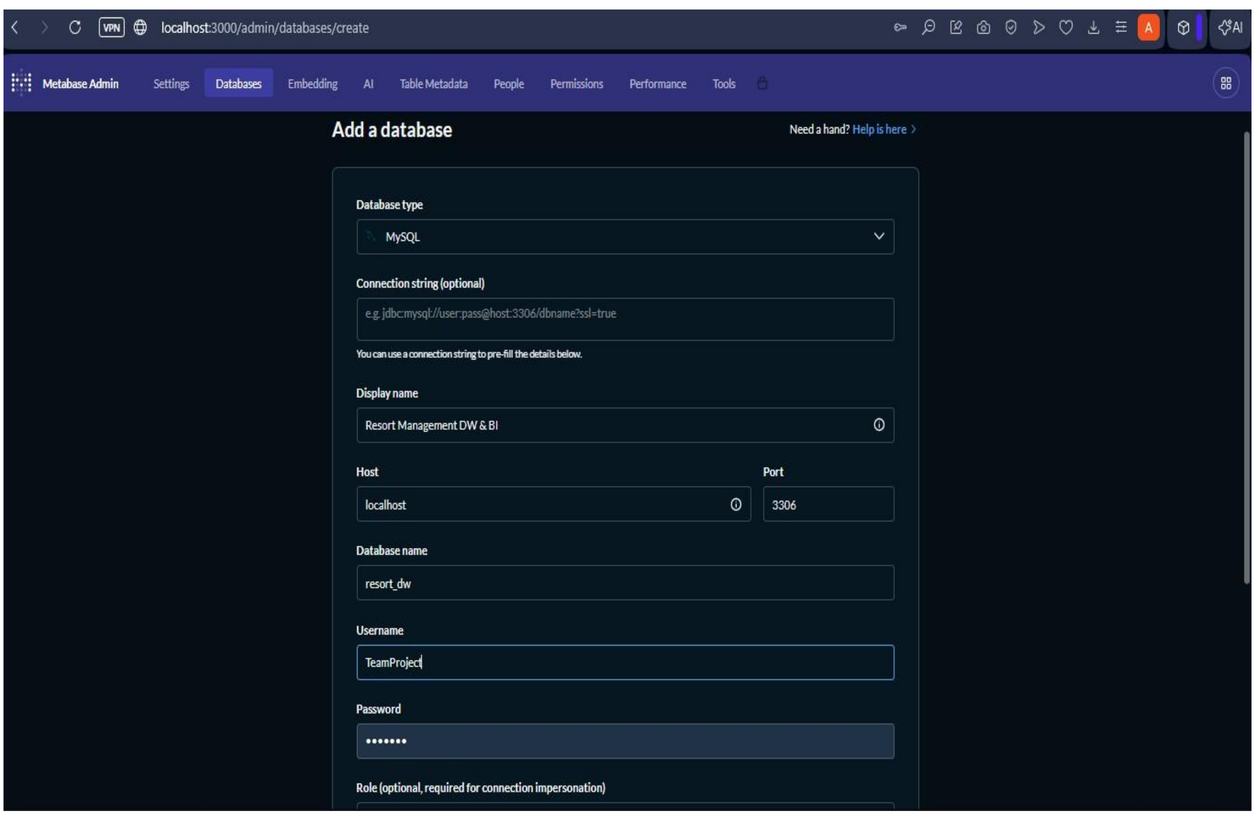


Figure 8.1: Metabase Data Source Configuration

Figure 8.2 shows the implemented Data Warehouse tables available within Metabase after successful connection.

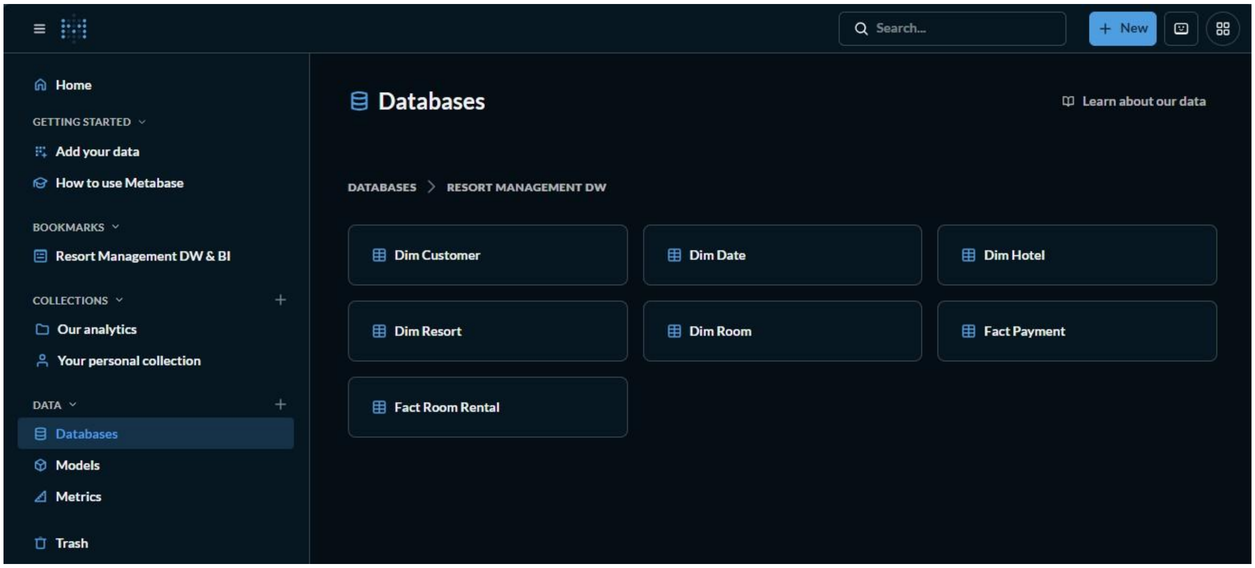


Figure 8.2: resort_dw Database in Metabase

The connection enabled direct access to the dimension and fact tables implemented in the Data Warehouse and provided the foundation for dashboard and report development.

8.3 Executive Dashboard

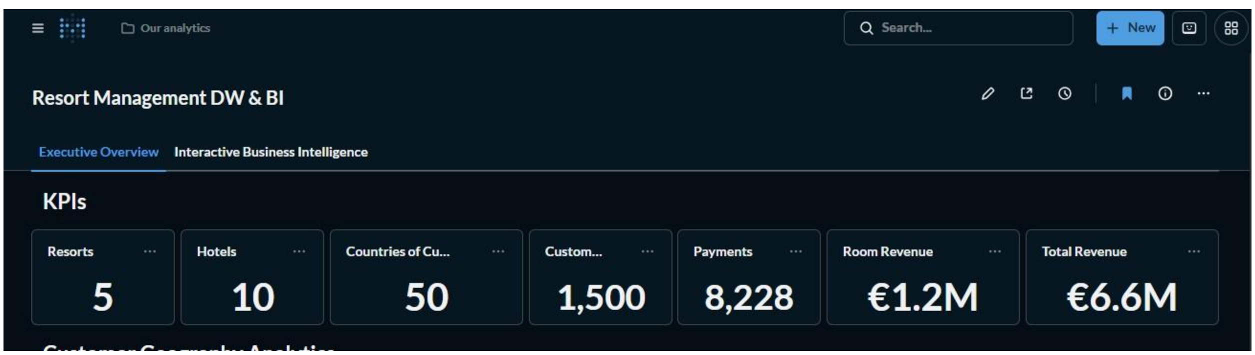


Figure 8.3: Executive Dashboard

The Executive Dashboard provides a high-level overview of resort operations through KPI cards and summary visualizations.
The dashboard includes the following KPI cards:

- Total Resorts
- Total Hotels

- Customer Countries
- Total Customers
- Total Payments
- Room Revenue
- Total Revenue

In addition to KPI monitoring, the dashboard contains visualizations grouped into the following analytical categories:
- Customer Geography Analytics
- Customer Demographics Analytics
- Season and Time Analytics
- ROOM_RENTAL Analytics

These visualizations provide a consolidated view of business performance and support executive-level decision-making.

8.4 Interactive Dashboard

Figure 8.4: Interactive Dashboard

The Interactive Dashboard was developed to support multidimensional exploration of Data Warehouse information.
Unlike the Executive Dashboard, the Interactive Dashboard includes filters that allow users to dynamically modify the displayed results.
The available filters include:
- Year
- Month
- Business Season
- Country
- Gender

- Age Group
- Resort
- Hotel
- Room Type

These filters are connected to the dashboard visualizations and KPI cards, allowing users to analyse information from different perspectives and levels of detail.
The Interactive Dashboard supports filter-driven multidimensional analysis across time, geography, customer demographics, accommodation characteristics, and organizational structures.

8.5 Analytical Reports

In addition to the dashboards, several analytical reports were developed to answer selected decisional queries and demonstrate the capabilities of the implemented Data Warehouse.

8.5.1 Revenue Analysis Report

As shown in Figure 8.5, the report directly supports **Query 1**, which examines revenue generated by customers from Italy during August 2025.

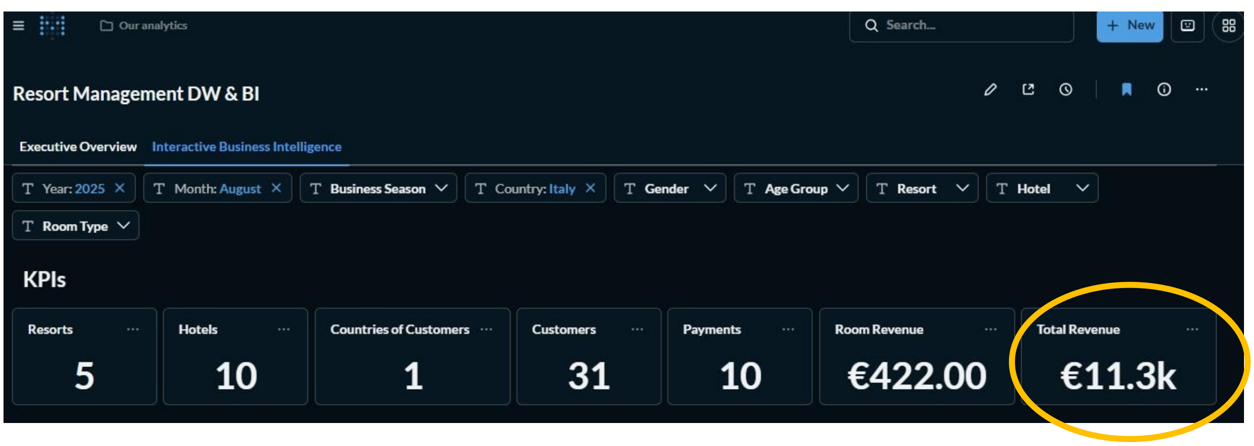


Figure 8.5: Revenue Analysis Report

The Revenue Analysis Report was developed using the Payment Data Mart.
The report analyses revenue across customer geography and time dimensions using measures derived from FACT_PAYMENT.

Managers can use this report to evaluate customer markets, identify revenue trends, and compare performance across different countries and seasons.

8.5.2 ROOM_RENTAL Analysis Report

As shown in Figure 8.6, the report directly supports **Query 3**, which focuses on total rental income generated by Suite room types within Hotel Alpha during the Summer Business Season of 2025.

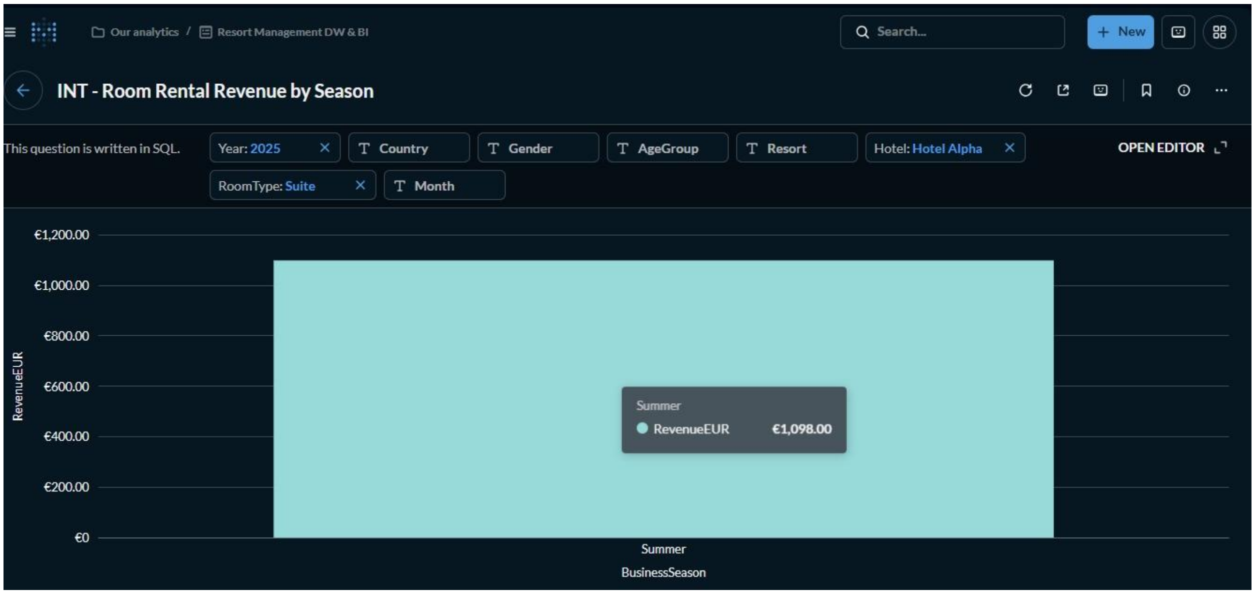


Figure 8.6: ROOM_RENTAL Analysis Report

The ROOM_RENTAL Analysis Report was developed using the ROOM_RENTAL Data Mart. The report analyses rental revenue and accommodation performance across room, hotel, resort, and time dimensions using FACT_ROOM_RENTAL.

Managers can identify room types that contribute significantly to revenue and evaluate accommodation performance across different locations.

8.5.3 Customer Demographic Analysis Report

The Customer Demographic Analysis Report was developed using the Payment Data Mart. The report analyses payment behaviour across customer demographic and geographical dimensions using FACT_PAYMENT.

As shown in Figure 8.7, the report directly supports **Query 8**, which examines the average payment amount for male customers aged 65 years and above from Italy during 2025.

Managers can examine payment patterns across different customer segments based on gender, age group, and country.

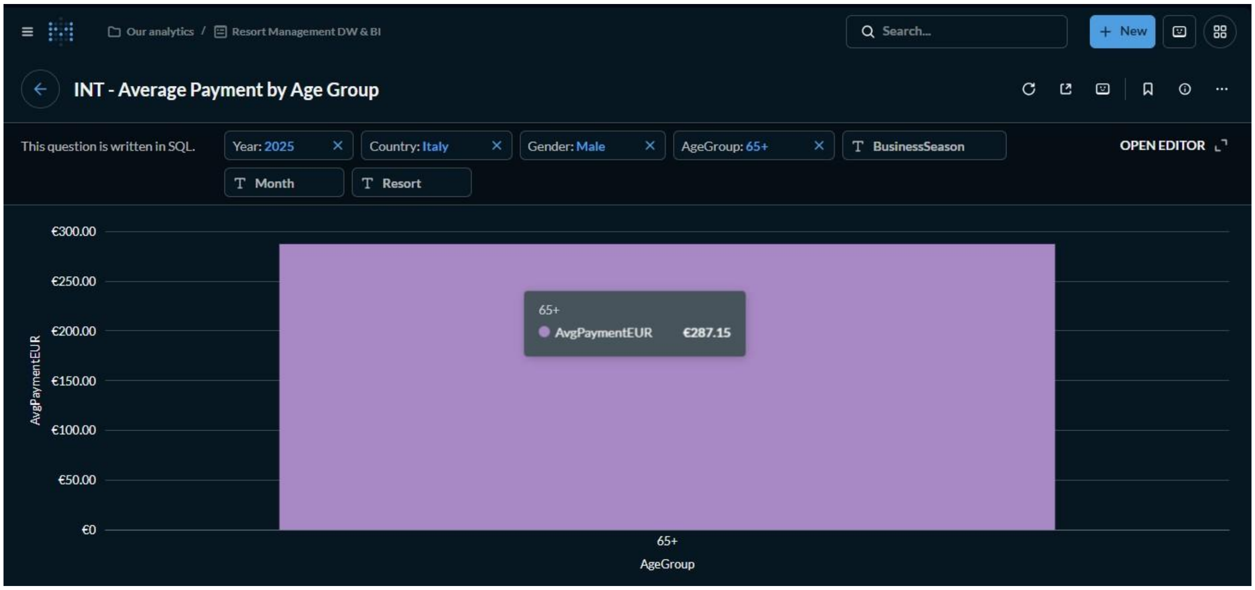


Figure 8.7: Customer Demographic Analysis Report

8.6 Decision Support Capabilities

The implemented Business Intelligence environment supports multidimensional analysis across revenue, accommodation performance, customer demographics, customer geography, and seasonal trends.
The Executive Dashboard provides a strategic overview of business performance, while the Interactive Dashboard supports detailed analytical exploration through dynamic filtering. Shared dimensions across the Payment and ROOM_RENTAL Data Marts ensure consistency in reporting and support integrated analysis across multiple business processes.

8.7 Summary

This chapter presented the Business Intelligence environment implemented using Metabase. KPI cards, analytical visualizations, interactive dashboards, and analytical reports were developed using the Data Warehouse as the analytical data source. The resulting environment supports multidimensional analysis and managerial decision-making across the resort management domain.

Chapter 9: Challenges and Lessons Learned

9.1 Overview

This chapter presents the main challenges encountered during the project and the lessons learned throughout the design and implementation of the Resort Management Data Warehouse and Business Intelligence solution.

9.2 Challenges Encountered

9.2.1 Schema Reification

One of the initial challenges was the transformation of the original E/R schema into a structure suitable for dimensional modelling. Several many-to-many relationships contained analytical information and required reification before conceptual modelling could begin.

9.2.2 Fact Selection

Multiple candidate facts were identified from the reified schema. Selecting the most suitable facts required analysing the decisional queries and determining which business processes provided the greatest analytical value. This resulted in the selection of PAYMENT and ROOM_RENTAL for implementation.

9.2.3 Dimensional Modelling

Designing appropriate dimensions, measures, and hierarchies required careful alignment with the decisional queries. Pruning and grafting were necessary to remove non-analytical attributes and introduce analytical hierarchies suitable for multidimensional analysis.

9.2.4 Data Generation and ETL

The operational database required a sufficiently large and realistic dataset for analytical processing. Generating consistent synthetic data while maintaining referential integrity across entities and relationships required additional validation during both data generation and ETL implementation.

9.2.5 Dashboard Design

Developing dashboards that effectively supported the selected decisional queries required balancing summary-level reporting with detailed analytical exploration. Appropriate KPIs, visualizations, and filters had to be selected to support different analytical perspectives.

9.3 Lessons Learned

The project provided practical experience in the complete Data Warehouse development lifecycle, from requirements analysis to Business Intelligence reporting.
Key lessons learned include:

- The importance of selecting facts based on business requirements.
- The role of reification in preparing operational schemas for analytical modelling.
- The significance of dimensions, hierarchies, and measures in supporting multidimensional analysis.
- The importance of ETL processes in transforming operational data into analytical structures.
- The value of dashboards and reports in supporting data-driven decision-making.

The project also demonstrated the relationship between Operational Databases, Data Warehouses, Data Marts, and Business Intelligence platforms within an integrated analytical environment.

9.4 Summary

This chapter presented the main challenges encountered during the project and the lessons learned throughout the implementation process. These experiences contributed to a deeper understanding of Data Warehouse design, ETL development, dimensional modelling, and Business Intelligence reporting.

Chapter 10: Conclusion and Future Work

10.1 Overview

This chapter concludes the Resort Management Data Warehouse and Business Intelligence project by summarizing the work completed, evaluating the objectives achieved, and identifying possible areas for future enhancement.

10.2 Project Summary

The project implemented a complete Data Warehouse and Business Intelligence solution based on the Resort Management E/R schema provided for the assignment.
The project began with requirements analysis and the identification of decisional queries. Candidate facts were identified, the E/R schema was refined through reification, and conceptual modelling was performed using attribute trees, pruning and grafting, dimensions, measures, and fact schemas.
The resulting designs were transformed into snowflake schemas and implemented within a MySQL-based Data Warehouse. A SQL-based ETL process was developed to transform operational data into analytical structures. Finally, Metabase was integrated with the Data Warehouse to provide dashboards and analytical reports that support managerial decision-making.

10.3 Objectives Achieved

The objectives defined for the project were successfully achieved.
Specifically:

- Decisional queries were identified from the Resort Management domain.
- Candidate facts were identified and analysed.
- The original E/R schema was refined through reification.
- Conceptual and logical designs were developed.
- The operational database was implemented and populated.
- The Data Warehouse was implemented using dimension and fact tables.
- ETL processes were developed and executed successfully.
- Data marts were implemented for PAYMENT and ROOM_RENTAL.
- Business Intelligence dashboards and analytical reports were developed using Metabase.

These outcomes demonstrate the successful implementation of the complete Data Warehouse development lifecycle.

10.4 Future Work

Several enhancements may be considered in future implementations:

- Implementation of additional data marts based on the remaining candidate facts.
- Integration with real operational resort data instead of synthetic datasets.

- Automation of ETL processes through scheduled execution.
- Deployment of the solution within a cloud-based environment.
- Development of predictive analytics and forecasting models.
- Expansion of the Business Intelligence environment with additional dashboards and analytical reports.

10.5 Closing Remarks

The implemented solution demonstrates how operational data can be transformed into meaningful analytical information through Data Warehouse technologies and Business Intelligence tools. The resulting environment provides a structured foundation for multidimensional analysis and supports evidence-based decision-making within the resort management domain.

Team Members Contribution

All team members actively participated throughout the project lifecycle and contributed to the successful completion of the work.

Activities including requirements analysis, decisional query development, schema reification, conceptual modelling, logical design, operational database implementation, Data Warehouse implementation, ETL development, dashboard creation, report preparation, testing, and presentation were carried out collectively through continuous planning, discussion, and coordination.

The team therefore considers the contributions of all members to be substantially equal.

References

Golfarelli, M., & Rizzi, S. (2009). *Data warehouse design: Modern principles and methodologies*. McGraw-Hill.

Inmon, W. H. (2005). *Building the data warehouse* (4th ed.). Wiley.

Metabase. (n.d.). *Metabase documentation*. https://www.metabase.com/docs

Appendix A: GitHub Repository Information

The complete implementation of the Resort Management Data Warehouse and Business Intelligence project is available in the project GitHub repository.
The repository contains all source code, database scripts, Data Warehouse scripts, dashboard exports, diagrams, and project documentation used during the implementation.

Repository URL:
https://github.com/FrancisOfosu/resort-management-dw-bi

Repository Structure:

```
Resort_Management_DW_BI/
├── Python_Data_Generation/
├── OLTP_Database/
├── Data_Warehouse/
├── Metabase_Reports/
├── Diagrams/
├── Documentation/
└── README.md
```

Appendix B: Sample Generated Data

```sql
SELECT 'DIM_CUSTOMER' AS TableName, COUNT(*) AS TotalRecords FROM DIM_CUSTOMER
UNION ALL
SELECT 'DIM_DATE', COUNT(*) FROM DIM_DATE
UNION ALL
SELECT 'DIM_RESORT', COUNT(*) FROM DIM_RESORT
UNION ALL
SELECT 'DIM_HOTEL', COUNT(*) FROM DIM_HOTEL
UNION ALL
SELECT 'DIM_ROOM', COUNT(*) FROM DIM_ROOM
UNION ALL
SELECT 'FACT_PAYMENT', COUNT(*) FROM FACT_PAYMENT
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

TableName	TotalRecords
DIM_CUSTOMER	1500
DIM_DATE	1086
DIM_RESORT	5
DIM_HOTEL	10
DIM_ROOM	100
FACT_PAYMENT	8228
FACT_ROOM_RENTAL	900

Appendix C: ETL Mapping Summary

This appendix summarizes the mapping between the Operational Database and the Data Warehouse during the ETL process.

Table C.1: Dimension Mapping

Operational Source	Data Warehouse Target
PERSON	DIM_CUSTOMER
RESORT	DIM_RESORT
HOTEL	DIM_HOTEL
ROOM	DIM_ROOM
PAYMENT.Date	DIM_DATE
ROOM_RENTAL.Date	DIM_DATE

Table C.2: Fact Mapping

Operational Source	Data Warehouse Target
PAYMENT	FACT_PAYMENT
ROOM_RENTAL	FACT_ROOM_RENTAL

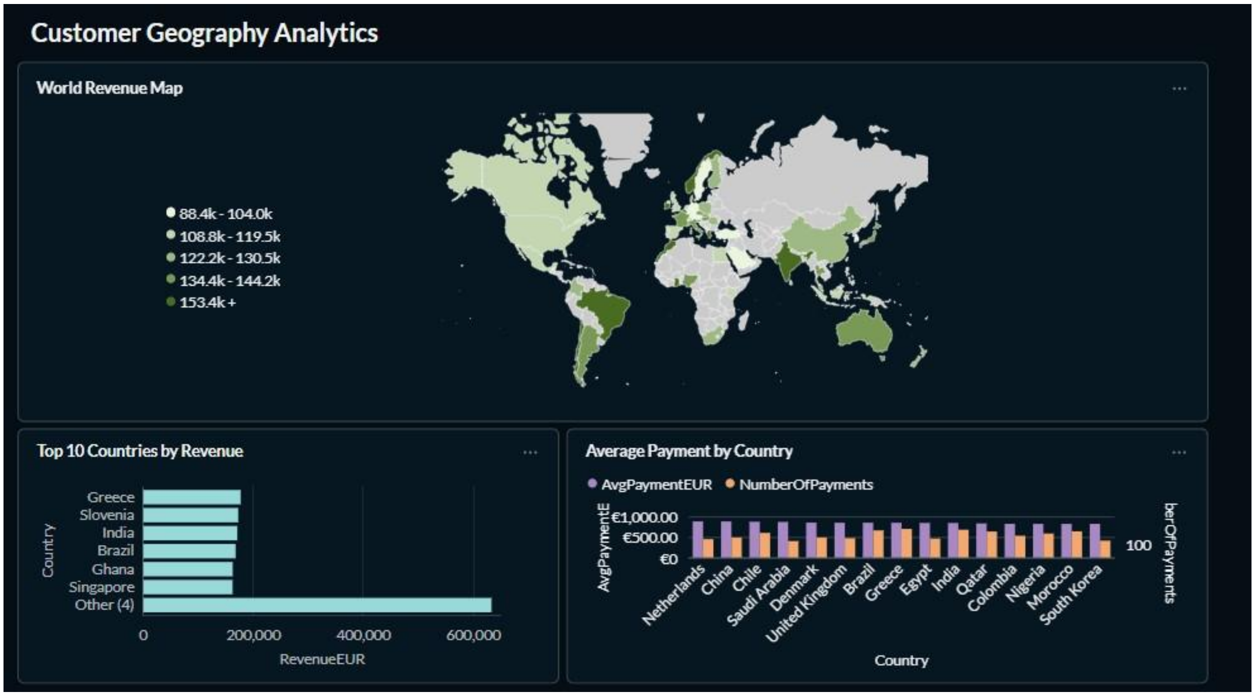
Table C.3: Derived Analytical Attributes

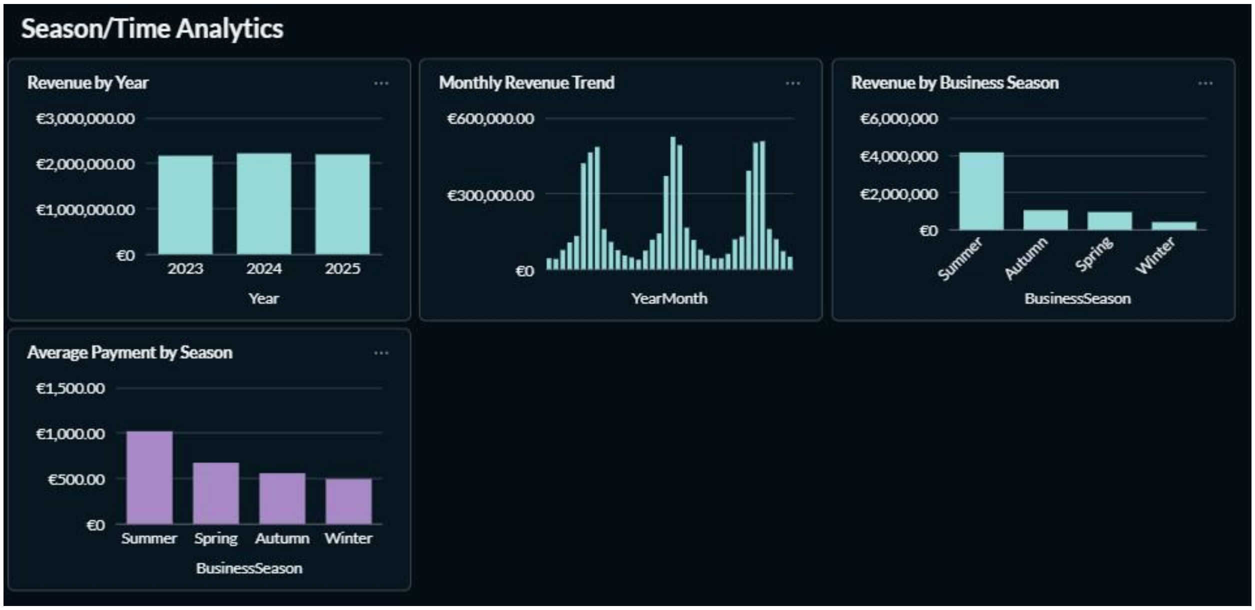
Data Warehouse Attribute	Source	Transformation
Business Season	Date	Derived from month
Day of Week	Date	Derived from transaction date
Month Number	Date	Derived from transaction date
Month Name	Date	Derived from transaction date
Quarter Number	Date	Derived from transaction date
Year Number	Date	Derived from transaction date
Age Group	PERSON.Age	Categorization into age ranges
RentalDays	ROOM_RENTAL.Period	Conversion to numerical duration
RentalRevenue	ROOM_RENTAL, ROOM	Room Price × RentalDays

Appendix D: Additional Dashboard Screenshots

This appendix contains supplementary dashboard and report screenshots that were not included in Chapter 8.

The screenshots provide additional examples of the analytical capabilities implemented within the Metabase Business Intelligence environment.




Season/Time Analytics
Revenue by Year
€3,000,000.00
€2,000,000.00
€1,000,000.00
€0
2023
2024
2025
Year
Monthly Revenue Trend
€600,000.00
€300,000.00
€0
YearMonth
Revenue by Business Season
€6,000,000
€4,000,000
€2,000,000
€0
Summer
Autumn
Spring
Winter
BusinessSeason
Average Payment by Season
€1,500.00
€1,000.00
€500.00
€0
Summer
Spring
Autumn
Winter
BusinessSeason


Room Rental Analytics
Revenue by Room Type
€300,000.00
€200,000.00
€100,000.00
€0
Deluxe
Family
Suite
Double
Single
RoomType
Rental Count by Room Type
300
200
100
0
Double
Family
Suite
Single
Deluxe
RoomType
Revenue by Hotel
HotelName
Sea View Hotel
Hotel Europa
Central Comfort Hotel
Royal Garden Hotel
Golden Horizon Hotel
Urban Stay Hotel
Other (4)
€0
RevenueEUR
Revenue by Resort
€300,000.00
€200,000.00
€100,000.00
€0
Sunset Bay Resort
Ocean Breeze Resort
Royal Palm Resort
Lake Paradise Resort
Mountain View Resort
ResortName
Average Length of Stay by Room Type
6
3
0
Family
Single
Double
Suite
Deluxe
RoomType
Room Rental Revenue by Season
€600,000.00
€300,000.00
€0
Summer
Autumn
Spring
Winter
BusinessSeason
Average Revenue per Rental by Room T...
€3,000.00
€2,000.00
€1,000.00
€0
Deluxe
Suite
Family
Double
Single
RoomType